Dr. Ragadeepika Pucha

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POSITIONS

- **Postdoctoral Researcher** **2023 – Present**

Department of Physics & Astronomy, University of Utah

EDUCATION

- **Ph.D in Astronomy and Astrophysics** **2016 – 2023**

Steward Observatory, University of Arizona
Thesis: *Dwarf Galaxies As Probes of Galaxy Evolution*
Advisors: Stéphanie Juneau, Arjun Dey

- **M.S in Astronomy and Astrophysics** **2016 – 2018**

Steward Observatory, University of Arizona
Advisors: Beth Willman, Jeffrey Carlin

- **Five Year Integrated M.Sc (Physics)** **2010 – 2015**

Integrated Science Education and Research Center (ISERC)
Visva-Bharati University, India

SELECTED AWARDS AND HONOURS

- **2026: College of Science Outstanding Postdoctoral Researcher Award**, University of Utah.
- **2025: Outstanding Postdoctoral Researcher Award**, Department of Physics & Astronomy, University of Utah.
- **2022: FAMOUS grant**, American Astronomical Society.
- **2022: Best Poster Award**, Poster Symposium Targeting Early-career Researchers (PoSTER) Conference.
- **2020: Jamieson Graduate Fellowship**, University of Arizona.
- **2016 – 2017: College of Science Fellowship**, University of Arizona.
- **2015: International Max Planck Research School (IMPRS) Fellowship**, Max Planck Institute for Solar System Research, Göttingen, Germany.
- **2012 – 2014:** Three consecutive years of **Innovation in Science Pursuit for Inspired Research** (INSPIRE) Summer Fellowship, Department of Science and Technology, India.
- **2010 – 2015: Innovation in Science Pursuit for Inspired Research** (INSPIRE) Fellowship for Undergraduate studies, Department of Science and Technology, India.

PUBLICATIONS

My ADS Library: I have authored 20+ refereed journal papers, including 5 first-author papers and contributed significantly to 2 other papers. See my complete list of publications at the end.

GRANTS AND PROPOSALS

- NASA ADAP Proposal, Science PI (PI: Prof. Yao-Yuan Mao)

Co-evolution of Dwarf Galaxies and Black Holes **2025 – 2027**

- Keck/Subaru PFS 2026B, Co-I (PI: Prof. Daniel Weisz)

Characterising Isolated Dwarf Galaxy Evolution with the Stellar Halo of IC 1613 **November 2026**

- Gemini Telescope Fast Turnaround Program 2024B, PI

Discovery of the Lowest-Mass Black Hole in a Dwarf Galaxy **September 2024**

- DESI Secondary Target Program, PI

Intermediate-Mass Black Holes in Low-Mass Galaxies **2021 – Present**

SCIENTIFIC COLLABORATIONS

Dark Energy Spectroscopic Instrument (DESI) Member **2019 – Present**

- Builder **2026 – Present**
- Galaxy & Quasar Physics AGN Topical Group Co-Lead **2024 – 2025**

TECHNICAL SKILLS

- Programming: Python (primary), IDL, C, C++, SQL, GitHub, Slurm supercomputing.
- Softwares: ds9, DRAGONS, GALFIT, GELATO, IRAF, Source Extractor, TOPCAT, LaTeX.
- Astronomy-related Python packages: astropy, astroconda, astrodrizzle, astroquery, lsdb.

- Astronomy-related Platforms: Astro Data Lab, Rubin Science Platform.
- `EmFit` - developed a Python-based spectral emission-line fitting code.

CONTRIBUTIONS TO PUBLICLY AVAILABLE DATA PRODUCTS

- `EmFit` Value-Added Catalogue: Emission-line fitting results for low-redshift ($z \leq 0.45$) galaxies, *Pucha, R. et al. 2025, Pucha et al. 2026*.
- Value-Added Catalogue of Physical Properties of DESI Galaxies: Stellar masses and other physical properties from spectral energy distribution modelling, including AGN templates, *Siudek, M., Pucha, R. et al., 2024*.
- AGN/QSO Summary Value-Added Catalogue: AGN and QSO Identifications for all targets in DESI, *DESI Galaxy and Quasar Physics Working Group including Pucha, R. et al., in preparation*.
- FastSpecFit Emission-line Catalogue: Spectrophotometric fitting results from the `FastSpecFit` stellar continuum and emission-line modelling code, *Moustakas, J., including Pucha, R. et al., in preparation*.

SELECTED TALKS AND POSTERS

Invited Talks

- **2026:** Hand-in-Hand: Rubin and Complementary Datasets Workshop, University of Utah, USA.
Combining Large Datasets with Astro Data Lab.
- **2025:** Galaxy Evolution Seminar, University of Oxford, UK.
Active Black Holes in Dwarf Galaxies: Insights from DESI.
- **2025:** AGN Coffee, ESO, Garching, Germany.
Unveiling Active Black Holes in Dwarf Galaxies with DESI.
- **2024:** DESI December Collaboration Meeting, Plenary Talk, Cancun, Mexico.
Tripling the Census of Dwarf AGN Candidates Using Early DESI Data.
- **2022:** CosmoPalooza, 239th AAS Meeting, Online.
Active Black Holes in Dwarf Galaxies with DESI Survey Validation.
- **2021:** Plenary Talk, DESI Collaboration Meeting, Online.
Active Galactic Nuclei in Low-mass Galaxies using DESI Survey Validation.
- **2021:** ISM* Series, Space Telescope Institute (STScI), Online.
Escape of Lyα in Lyman-Alpha Emitters: Dependence on Galaxy Properties and Environment.
- **2021:** Astro Data Lab Special Session, 237th AAS Meeting, Online.
Joint Spectroscopic and Photometric Analysis of Low-Redshift Galaxies.

Selected Contributed Talks

- **2026:** Empowering Science in the Data-Rich Era of Astronomy, NSF NOIRLab, Tucson, USA.
Scaling Up the Census of Active Black Holes in Dwarf Galaxies with DESI.
- **2025:** Massive Black Holes Across Cosmic Time, Kavli Institute for Cosmology, Cambridge, UK.
A Tenfold Increase in the Census of Dwarf AGN Candidates Using DESI DR1 Data.
- **2025:** Quo Vadis Galaxy Evolution?, University of Heidelberg, Heidelberg, Germany.
Exploring AGN and SF Feedback in Dwarf Galaxies: A DESI Perspective.
- **2025:** Galactic Frontiers II Conference, Dartmouth College, Hanover, USA.
Active Black Holes in Dwarf Galaxies: A DESI Perspective.
- **2023:** DESI December Collaboration Meeting, Hawaii, USA.
Multi-Component Emission-Line Fitting in Low-Redshift DESI Targets.
- **2023:** Coordinating the Next Generation of Spectroscopic Processing and Analysis Tools, NSF NOIRLab, Tucson.
Emission-Line Fitting Using Astropy Modelling.
- **2022:** What Drives the Growth of Black Holes?, Reykjavik, Iceland.
Black Hole Seeds in Dwarf Galaxies: Early Results from DESI.
- **2022:** Origin, Growth, and Feedback of Black Holes in Dwarf Galaxies conference, San Sebastian, Spain.
Co-evolution of Dwarf Galaxies and their Black Holes: Early Results from DESI.
- **2021:** Data Tutorial, DESI Collaboration Meeting, Online.
Introduction to DESI Everest Spectroscopic Data Release.

Selected Posters

- **2022:** Poster Symposium Targeting Early-career Researchers (PoSTER) Conference.
Lyman-Alpha Escape from Low-mass, Compact, High-Redshift Galaxies (best poster award).
- **2022:** Poster Symposium Targeting Early-career Researchers (PoSTER) Conference.
Active Black Holes in Dwarf Galaxies from DESI.

Research Student Supervision

- Abantika Maytra Oishee, Undergraduate student.

Hajee Mohammad Danesh Science & Technology University, Bangladesh **2026 – Present**

- Lindsay Taylor, Undergraduate Student.

University of Utah

Co-advisor with Prof. Yao-Yuan Mao **2026 – Present**

- David Ohlson, Graduate Student.

University of Utah

Co-advisor with Prof. Yao-Yuan Mao and Prof. Anil Seth **2025 – Present**

- David Fowles, Undergraduate Student.

University of Utah

Co-advisor with Prof. Yao-Yuan Mao **2024 – 2025**

- Swagatha Bera, Undergraduate Student.

Visva-Bharati University, India **2020 – Present**

SELECTED MEDIA COVERAGE

- **2026:** Astronomy, Active Black Holes are More Common than We Thought

quoting me on the significance of a recent census of AGN candidates.

- **2025:** NOIRLab Press Release, *DESI Uncovers 300 New Intermediate-Mass Black Holes Plus 2500 New Active Black Holes in Dwarf Galaxies*

featuring my research on the discovery of AGN candidates in dwarf galaxies.

- **2025:** The University of Utah News, *'Vast discovery' of Black Holes in Dwarf Galaxies*

featuring my research on the discovery of AGN candidates in dwarf galaxies.

- **2025:** Space, *Largest-ever discovery of 'missing link' black holes revealed by dark energy camera*

featuring my research on the discovery of AGN candidates in dwarf galaxies.

- **2024:** ScienceNews, *A Cosmic Census Triples the Known Number of Black Holes in Dwarf Galaxies*

featuring my research on the discovery of AGN candidates in dwarf galaxies.

SELECTED COMMUNITY ACTIVITIES

- **2025:** *Tripling the Census of Active Black Holes in Dwarf Galaxies*, Public Talk, Utah Astronomy Club.
- **2019 – Present:** *Project Bharati:* An initiative for developing scientific interest among high-school and undergraduate students in India.
- **2021:** *Dwarf Galaxies as Probes of the Universe*, ISERC I/O online talk series, Visva-Bharati University.
- **2020:** *Importance of Women Education in India*, online workshop, Girl Up Bombay.
- **2020:** *Women in STEM*, online talk series, Stem4Gils, New Delhi.

SELECTED TEACHING AND MENTORING ACTIVITIES

- **2023 – Present:** *Mentor* to graduate students, DESI Mentorship Program.
- **2019 – Present:** *Mentor* to high-school and undergraduate students in India, Project Bharati.
- **2017 – 2022:** *Graduate Teaching Assistant*, University of Arizona
  - ASTR 202: Life in the Universe (Spring 2022)
  - ASTR 250: Fundamentals of Astrophysics (Fall 2020)
  - ASTR 203: Stars (Fall 2019, Fall 2021)
- **2018:** Tutor, *Astronomy Tutoring for Major and Minors Program* (ATOMM), University of Arizona.

REFERENCES

- Dr. Yao-Yuan Mao, Assistant Professor, University of Utah, *yymao@astro.utah.edu*
- Dr. Stéphanie Juneau, Associate Astronomer, NSF NOIRLab, *stephanie.juneau@noirlab.edu*
- Dr. Arjun Dey, Astronomer, NSF NOIRLab, *arjun.dey@noirlab.edu*

LIST OF PUBLICATIONS

My ADS Library

Major Contributions

- **Pucha, R.**, Juneau, S., Mezcua, M. et al., **2026**, ApJ, 1009, 150.
A New Record Census of Dwarf AGN and a Bimodal $M_{BH} - M_{\star}$ Scaling Relation with DESI DR1.
- Morales, V., Mezcua, M., **Pucha, R.** et al., Submitted to MNRAS
The Largest Sample of AGN Outflows in Dwarf Galaxies using DESI DR1.
- **Pucha, R.**, Juneau, S., Dey, A., et al., **2025**, ApJ, 982, 10.
Tripling the Census of Dwarf AGN Candidates using DESI Early Data.
- Siudek, M., **Pucha, R.**, M., Mezcua, M. et al., **2024**, A&A, 691, A308.
Value-Added Catalog of Physical Properties of more than 1.3 million galaxies from the DESI Survey.
- **Pucha, R.**, Reddy, N., Dey, A., et al., **2022**, AJ, 164, 159.
Lyα Escape from Low-mass, Compact, High-Redshift Galaxies.
- **Pucha, R.**, Carlin, J., Willman, B., et al., **2019**, ApJ, 880, 104.
Hyper Wide Field Imaging of the Local Group Dwarf Irregular Galaxy IC 1613: An Extended Component of Metal Poor Stars.
- **Pucha, R.**, Hiremath, K.M., Gurumath, S., **2016**, Journal of Astrophysics and Astronomy, 37, 3.
Development of a Code to Analyze the Solar White-Light Images from the Kodaikanal Observatory: Detection of Sunspots, Computation of Heliographic Coordinates and Area.

Minor Contributions

- Alfarsy, R., Canning, R., Mueller, E., including **Pucha, R.**, et al., Submitted to MNRAS.
CIV Line Shape Analysis and Black Hole Mass Estimates of DESI Quasars.
- Jewell, S., Smethurst, R., Lintott, C., including **Pucha, R.**, et al., MNRAS, 552, 1783.
Galaxy–SMBH co-evolution in 2,435 DESI disk galaxies and merger-free BH growth in bulgeless disks.
- Arévalo, P., Buchner, J., Salvato, M., including **Pucha, R.**, et al., Submitted to MNRAS.
The X-ray-Weak Tail of the Quasar Population: Physical Properties and Redshift Evolution of Variability-Selected AGN in eFEDs.
- Dadiani, E., Palmese, A., Zhou, Y., including **Pucha, R.**, et al., Submitted to AAS Journals.
Cosmic Pairs: A DESI Census of Dual and Offset AGN as Precursors to Massive Black Hole Binaries
- Alcolea, J., Siudek, M., Eriksen, M., including **Pucha, R.**, et al., **2026**, A&A, 713, 133.
Beyond Traditional Emission-line Diagnostics: Using Autoencoders to uncover Active Galactic Nuclei in DESI Spectra.
- Alfarsy, R., Canning, R., Mueller, E., including **Pucha, R.**, et al., **2026**, MNRAS, 551, 1418.
Stacked Reverberation Mapping of High Redshift Quasars in DESI. I. Feasibility Analysis.
- Mezcua, M., Laloux, B., Scialpi, M., including **Pucha, R.**, et al., **2026**, A&A, 712, 83.
Euclid: Quick Data Release (Q1) - Dual AGN in low-mass galaxies
- DESI Collaboration, including **Pucha, R.**, et al., **2026**, AJ, 171, 285.
Data Release 1 of the Dark Energy Spectroscopic Instrument.
- Juneau, S., Jacques, A., Pothier, S., including **Pucha, R.**, et al., **2025**, ASPC, 541, 77.
SPARCL: SPectra Analysis and Retrievable Catalog Lab.
- de los Reyes, M., Asali, Y., Wechsler, R., including **Pucha, R.**, et al., **2025**, ApJ, 989, 91.
Stellar Mass Calibrations for Local Low-Mass Galaxies.
- Juneau, S., Canning, R., Alexandar, D., **Pucha, R.**, et al., **2025**, AJ, 169, 157.
Identifying Missing Quasars from the DESI Bright Galaxy Survey.
- DESI Collaboration, including **Pucha, R.**, et al., **2024**, AJ, 168, 58.
The Early Data Release of the Dark Energy Spectroscopic Instrument.
- DESI Collaboration, including **Pucha, R.**, et al., **2024**, AJ, 167, 62.
Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument.
- Fawcett, V., Alexander, D., Brodzeller, A, including **Pucha, R.**, et al., **2023**, MNRAS, 525, 5575.
A Striking Relationship between Dust Extinction and Radio Detection in DESI QSOs: Evidence for a Dusty Blow-out Phase in Red QSOs.
- DESI Collaboration, including **Pucha, R.**, et al., **2022**, AJ, 164, 207.
Overview of the Instrument for the Dark Energy Spectroscopic Instrument.

- Hviding, R., Hainline, K., Rieke, M., including **Pucha, R.**, et al., **2022**, AJ, 163, 224.
A New Infrared Criterion for Selecting Active Galactic Nuclei to Lower Luminosities.
- Nidever, D., Dey, A., Fasbender, K., including **Pucha, R.**, et al., **2021**, AJ, 161, 192.
Second Data Release of the All-sky NOIRLab Source Catalog.
- Pat, F., Juneau, S., Böhm, V., **Pucha, R.**, et al., **2020**, ASP Conference Series, Vol. 525, 67.
Reconstucting and Classifying SDSS DR16 Galaxy Spectra with Machine-Learning and Dimensionality Reduction Algoriths.
- Carlin, J., Garling, C., Peter, A., including **Pucha, R.**, et al., **2019**, ApJ, 886, 109.
Tidal Destruction in a Low-mass Galaxy Environment: The Discovery of Tidal tails around DDO 44.
- Schindler, J., Fan, X., McGreer, I., including **Pucha, R.**, et al., **2018**, ApJ, 863, 144.
The Extremely Luminous Quasar Survey in the Sloan Digital Sky Survey Footprint. II. The North Galactic Cap Sample.

Contributed Data Tutorial Jupyter Notebooks

- *DESI Data Tutorials*.
- *Introduction to DESI First Data Release (DR1) at the Astro Data Lab*.
- *Introduction to DESI Early Data Release (EDR) at the Astro Data Lab*.
- *Comparison of Spectroscopy from Sloan Digital Sky Survey (SDSS) and Dark Energy Spectroscopic Instrument (DESI) Survey*.
- *Multi-wavelength Image Cutouts and SDSS Spectra of Active Galaxies with Extreme Emission-Line Ratios*.
- *Stacking SDSS Spectra of Galaxies Selected from the BPT Emission-Line Diagnostic Diagram*.